Elementary matrices

Example

(a)
$$\begin{bmatrix} k & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} a & d & f \\ b & e & g \end{bmatrix}$$

$k \neq 0$

$$= \begin{bmatrix} ka & kd & kf \\ b & e & g \end{bmatrix}$$

(b)
$$\begin{bmatrix} 1 & 0 & k \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a & d \\ b & e \\ c & f \end{bmatrix}$$

$i = 1$
$j = 3$

$$= \begin{bmatrix} a+kc & d+kf \\ b & e \\ c & f \end{bmatrix}$$

(c) $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} a & d \\ b & e \\ c & f \end{bmatrix}$

$$= \begin{bmatrix} c & f \\ b & e \\ a & d \end{bmatrix}$$

These examples show that row operations can be obtained by multiplying on the left by certain square invertible matrices

Def$^{\underline{n}}$ An **invertible matrix** A is **elementary** if it has one of the following forms

(i) $k \neq 0$

$$\begin{array}{cc} & \text{i-th column} \\ \text{i-th row} & \begin{bmatrix} 1 & & & & & \\ & \ddots & & & & \\ & & 1 & & & \\ & & & k & & \\ & & & & 1 & \\ & & & & & \ddots & \\ & & & & & & 1 \end{bmatrix} \end{array} = C_i(k)$$

[Multiplying by this on the left scales the i-th row by k]

(ii) $i < j$, $k \in F$

$$\begin{array}{cc} & \text{j-th column} \\ \text{i-th row} & \begin{bmatrix} 1 & & 0 & & k & 0 \\ & 1 & & & & \\ & & \ddots & & & 0 \\ & 0 & & \ddots & & \\ & & & & \ddots & 1 \\ 0 & & 0 & & & \end{bmatrix} \end{array} = T_{i,j}(k)$$

$i > j$

$k \in F$

$T_{i,j}(k)$

$$\begin{array}{cc}
\begin{bmatrix}
1 & & & & 0 \\
 & \ddots & & & \\
0 & & \ddots & & 0 \\
 & & & \ddots & \\
k & & & & \\
0 & & 0 & & 1
\end{bmatrix} & \begin{array}{c} \\ \\ \\ \\ i\text{-th row} \\ \\ \end{array} \\
\begin{array}{l} \uparrow \\ j\text{-th column} \end{array} &
\end{array}$$

[Multiplying by this matrix on the left adds k-times the j-th row to the i-th row]

(iii)

$i < j$

$P_{i,j}$

$$\begin{array}{cc}
\begin{array}{ccccc} & i\text{-th column} & & j\text{-th column} & \end{array} & \\
\begin{bmatrix}
1 & & 0 & & 0 \\
 & 1 & & & \\
 & 0 & & 1 & \\
0 & & 1 & & 0 \\
 & 1 & & 0 & \\
 & & & & 1 \\
0 & & 0 & & 1 \\
 & & & & 1
\end{bmatrix} & \begin{array}{c} \\ \\ i\text{-th row} \\ \\ j\text{-th row} \\ \\ \\ \\ \end{array}
\end{array}$$

[Multiplying by this matrix on the

left switches the i-th & j-th rows]

Observation:

① $B \in M_{n\times m}(F)$

There exist elementary matrices

$E_1, E_2, \ldots, E_r \in M_{n\times n}(F)$

s.t. $E_1 E_2 \ldots E_r B$ is in RREF

② If $B \in M_{n\times n}(F)$ is invertible

then $E_1 E_2 \ldots E_r B = I_n$

$\Leftrightarrow E_1 E_2 \ldots E_r = B^{-1}$

$\Leftrightarrow B = (E_1 \ldots E_r)^{-1}$

$= E_r^{-1} E_{r-1}^{-1} \ldots E_1^{-1}$

(3) $E \in M_{n\times n}(F)$ elementary

$\Rightarrow$ $E^{-1} \in M_{n\times n}(F)$ is also elementary & of the same type

(i) $C_i(k)^{-1} = C_i(k^{-1})$

(ii) $T_{i,j}(k)^{-1} = T_{i,j}(-k)$

(iii) $P_{i,j}^{-1} = P_{i,j}$

(4) (2) + (3) $\Rightarrow$ Every invertible matrix is a product of elementary matrices

(5) $E \in M_{n\times n}(k)$ elementary then E^T is also elementary

(i) $C_i(k)^T = C_i(k)$

(ii) $T_{i,j}(k)^T = T_{j,i}(k)$

(iii) $P_{i,j}^T = P_{i,j}$

Proposition

$$\det(A^T) = \det(A)$$

Pf: Only has content if A is invertible

$$A = \underbrace{E_1 E_2 \ldots E_r}_{\text{elementary}} \quad (*)$$

$$\Rightarrow A^T = E_r^T E_{r-1}^T \cdots E_1^T \quad (**)$$

Prop ② from Lecture 24

$(*) \Rightarrow \det(A) = \det(E_1)\det(E_2) \cdots \det(E_r)$

$(**) \Rightarrow \det(A^T) = \det(E_r^T)\det(E_{r-1}^T) \cdots \det(E_1^T)$

So to show $\det(A) = \det(A^T)$

it's enough to consider the case where A is elementary

We can now check this directly for each of the cases (i), (ii) & (iii)

Example

$$\overset{A}{\begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix}}\overset{B}{\begin{bmatrix}5 & 6\\ 7 & 8\end{bmatrix}}$$

$$= \begin{bmatrix}19 & 22\\ 43 & 50\end{bmatrix} = AB$$

transpose

$$\begin{bmatrix}1 & 3\\ 2 & 4\end{bmatrix}\begin{bmatrix}5 & 7\\ 6 & 8\end{bmatrix}$$

$$= \begin{bmatrix}23 & \\ & \end{bmatrix}$$

$$\overset{B^T}{\begin{bmatrix}5 & 7\\ 6 & 8\end{bmatrix}}\overset{A^T}{\begin{bmatrix}1 & 3\\ 2 & 4\end{bmatrix}}$$

$$(AB)^T$$

$$\|$$

$$= \begin{bmatrix}19 & 43\\ 22 & 50\end{bmatrix} = B^T A^T$$

Observation 6

$$(\underbrace{A}_{m\times p}\underbrace{B}_{p\times n})^T = \underbrace{B^T}_{n\times p}\underbrace{A^T}_{p\times m}$$

$$B = \begin{bmatrix} v_1 & v_2 & \cdots & v_n \end{bmatrix} \qquad A = \begin{bmatrix} w_1^T \\ \vdots \\ w_m^T \end{bmatrix}$$

$$AB = \begin{bmatrix} w_i \cdot v_j \end{bmatrix}$$

← transpose

$$A^T = \begin{bmatrix} w_1 & \cdots & w_m \end{bmatrix} \qquad B^T = \begin{bmatrix} v_1^T \\ v_2^T \\ \vdots \\ v_n^T \end{bmatrix}$$

$$B^T A^T = \begin{bmatrix} v_i \cdot w_j \end{bmatrix}$$